

|          |                         |        |      |        |          |          |          |
|----------|-------------------------|--------|------|--------|----------|----------|----------|
| RELEASE  | v1.2.0                  | PYTHON | 3.8+ | NMAP   | REQUIRED | NETEXEC  | OPTIONAL |
| PLATFORM | LINUX   MACOS   WINDOWS |        |      | STATUS | ACTIVE   | SECURITY | VAPT     |

# scanrunner

## v1.2.0 Detailed User Manual

Nmap + NetExec Workflow Automation for Authorized Assessments



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Manual edition: 19 July 2026

### **AUTHORIZED USE ONLY**

Use scanrunner only against systems you own or are explicitly authorized to assess.

## Legal and Safety Notice

### Authorization requirement

scanrunner orchestrates network discovery and security-enumeration tools. Scanning systems without explicit authorization can violate law, contract, acceptable-use policies, or client rules of engagement. Always define and verify scope before starting.

- Obtain written authorization and a defined testing window.
- Use `--scope-file` to enforce engagement boundaries where practical.
- Confirm whether intrusive Nmap NSE scripts, UDP scans, OS detection, or NetExec authentication checks are permitted.
- Protect output files because they may contain hostnames, services, operating-system details, usernames, or other sensitive information.
- Review generated commands before running them in production environments.

## How to Use This Manual

Read Chapters 1-4 for setup and first use. Use the workflow chapters during assessments, and keep the cheat sheets and troubleshooting chapters nearby for quick reference.

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# 1. Product Overview

scanrunner is a Python-based orchestration layer for Nmap and NetExec. It preserves the flexibility of the underlying tools while adding target management, reporting, resume logic, safety controls, and a focused command-line interface.

## 1.1 What scanrunner does

- Accepts a target file or a single IP, hostname, URL-style target, or CIDR.
- Passes unknown scan arguments directly to Nmap.
- Provides named Nmap templates for common assessment tasks.
- Detects completed and incomplete reports.
- Supports interactive decisions and non-interactive automation.
- Runs bounded parallel Nmap workers.
- Integrates NetExec protocols and creates focused tables.
- Writes Nmap normal output, XML, CSV, JSON, optional HTML, and audit logs.
- Enforces an optional scope allowlist.

## 1.2 What scanrunner does not do

- It does not replace Nmap, NetExec, or professional assessment judgment.
- It does not grant authorization.
- It does not guarantee that a host is secure when no open ports or findings are observed.
- It does not automatically interpret every Nmap or NetExec result as a vulnerability.
- It does not remove the need to validate false positives and environmental impact.

## 1.3 Main operating modes

| Mode          | Purpose                                                                     | Typical invocation                          |
|---------------|-----------------------------------------------------------------------------|---------------------------------------------|
| Nmap          | Host discovery, port scanning, service detection, NSE scripts, OS detection | scanrunner -f targets.txt -sV -T4           |
| Nmap template | Apply a predefined Nmap argument set                                        | scanrunner -f targets.txt --template quick  |
| NetExec       | Run an installed NetExec protocol                                           | scanrunner -f targets.txt -nxc smb          |
| Split         | Divide a target file into balanced parts                                    | scanrunner -f targets.txt --split 3         |
| Reporting     | Produce inventory and optional HTML output                                  | scanrunner -f targets.txt --html-report -sV |

## 2. Architecture and Execution Flow

The wrapper separates input preparation, validation, scanner execution, report inspection, and inventory generation. Understanding this flow makes fast or unexpected results easier to diagnose.

### High-level execution flow

```
User command
|
+-- Early help/version routing
|
+-- Parse scanrunner options
|
+-- Normalize and validate targets
|
+-- Enforce scope boundaries
|
+-- Select mode
    |
    +-- Nmap: report check -> ping/discovery decision -> scan -> validate -> inventory
    |
    +-- NXC: prepare target source -> execute NetExec -> parse -> table/CSV/JSON
    |
    +-- Split: deduplicate -> balance -> write part files
```

### 2.1 Nmap command construction

```
nmap [--stats-every 15s] [optional -Pn] [template/raw Nmap args]
-oN results/<target>.txt
-oX results/<target>.xml
-- <target>
```

#### Why the final -- is present

The double dash ends option parsing before the target. This reduces ambiguity when a target string could otherwise resemble an option.

### 2.2 NetExec command construction

```
nxc <protocol> <target-or-cleaned-target-file> [native NetExec options]
```

#### Credential display safety

Sensitive command values such as passwords, hashes, and keys are redacted in the displayed command. The real values are still passed to the scanner process.

## 3. Installation and Upgrade

The recommended installation uses `install.py`. It detects Linux, macOS, or Windows and configures a launcher and shell completion where supported.

### 3.1 Standard installation

```
git clone https://github.com/Madhav-Sai/scanrunner.git
cd scanrunner
python3 install.py
```

### 3.2 System-wide installation on Linux/macOS

```
sudo python3 install.py
```

#### Symlink behavior

The installer should create an absolute launcher such as `/usr/local/bin/scanrunner` -> `/home/user/scanrunner/scanrunner.py`. An absolute path prevents the broken-relative-symlink problem.

### 3.3 User-only installation

```
python3 install.py --user
```

- The launcher is placed under the current user, commonly `~/.local/bin`.
- Ensure `~/.local/bin` is in `PATH`.

### 3.4 Windows installation

```
py install.py
```

- The installer creates a user-level `scanrunner.cmd` launcher.
- PowerShell completion may be installed under the user profile.
- Open a new PowerShell session after `PATH` changes.

### 3.5 Installer options

| Option                                                     | Meaning                                  |
|------------------------------------------------------------|------------------------------------------|
| <code>--user</code>                                        | Install for the current user.            |
| <code>--system</code>                                      | Request a system-wide Unix installation. |
| <code>--shell auto</code>                                  | Detect the active shell.                 |
| <code>--shell zsh   bash   fish   powershell   none</code> | Force a completion target.               |

| Option                       | Meaning                                        |
|------------------------------|------------------------------------------------|
| <code>--no-completion</code> | Install the launcher without shell completion. |

## 3.6 Dependencies

```
# Debian / Ubuntu / Kali
sudo apt update
sudo apt install nmap

# macOS
brew install nmap
```

- Nmap is required for Nmap workflows.
- NetExec is optional and required only for `-nxc` mode.

## 3.7 Verify installation

```
which scanrunner
readlink -f "$(which scanrunner)"
scanrunner -v
scanrunner -h
nmap --version
nxc --version
```

## 3.8 Upgrade

```
cd ~/scanrunner
git pull
python3 -m py_compile scanrunner.py install.py
python3 -m unittest discover -s tests -v
sudo python3 install.py
hash -r
```

### Release tags

For a production assessment workstation, consider checking out a known release tag instead of following main continuously.

# 4. First Run and Help System

The CLI intentionally uses nested help so users see only the workflow they need.

## 4.1 Top-level help

```
scanrunner -h
```

## 4.2 Version

```
scanrunner -v
scanrunner --version
```

### Nmap verbosity remains available

Standalone scanrunner -v shows the scanrunner version. When a target is supplied, -v can be passed through to Nmap, for example scanrunner -i 10.10.10.10 -v -sV.

## 4.3 Focused help pages

| Command                  | Displays                                     |
|--------------------------|----------------------------------------------|
| scanrunner --nmap -h     | Nmap workflow and pass-through behavior.     |
| scanrunner --template -h | Template names, aliases, and arguments.      |
| scanrunner -nxc -h       | scanrunner NetExec mode and focused queries. |
| scanrunner --split -h    | Target-file splitting.                       |
| scanrunner --reports -h  | Reporting and automation.                    |
| scanrunner --help-all    | The complete reference.                      |

## 4.4 Option-specific help

```
scanrunner --parallel -h
scanrunner --help output
scanrunner --help scope-file
```

# 5. Target Input and Normalization

Every scan starts with either a target file (-f) or a single target (-i). These options are mutually exclusive.

## 5.1 Single target

```
scanrunner -i 10.10.10.10 -sV
scanrunner -i dc01.corp.local -sV
scanrunner -i 10.10.10.0/24 -sV
```

## 5.2 Target file

```
# targets.txt
10.10.10.10
10.10.10.20      # application server
dc01.corp.local
https://portal.example.com/login
10.10.20.0/24
```

```
scanrunner -f targets.txt -sV -T4
```

## 5.3 Normalization behavior

| Input                         | Normalized target |
|-------------------------------|-------------------|
| https://example.com/login     | example.com       |
| http://example.com:8080/admin | example.com       |
| example.com:443               | example.com       |
| 10.10.10.10                   | 10.10.10.10       |
| 10.10.10.0/24                 | 10.10.10.0/24     |

## 5.4 Deduplication and comments

- Blank lines are ignored.
- Lines beginning with # are ignored.
- Inline text after # is stripped.
- Duplicate normalized targets are retained only once.

## 5.5 Hostnames and DNS

### Hostname scans

A hostname must resolve through the scanner system DNS configuration or local hosts file. A DNS issue can look like a scan that is stuck or finishes unexpectedly. Test with `getent hosts <name>`, `host <name>`, or `dig <name>`.

## 6. Nmap Workflow

In Nmap mode, scanrunner manages one report per target while preserving ordinary Nmap syntax.



## 6.1 Raw Nmap arguments

```
scanrunner -i 10.10.10.10 -sV -sC -T4
scanrunner -i 10.10.10.10 -sS -p- -T4
scanrunner -f targets.txt --script vuln -sV
scanrunner -f web.txt -p 80,443,8080,8443 --script http-title,http-headers
```

## 6.2 Pass-through model

Arguments not recognized as scanrunner options are passed directly to Nmap. This allows use of Nmap options and NSE scripts without scanrunner needing to reimplement them.

### Quoted arguments

Keep multi-word Nmap arguments quoted. Example: `--script-args "user=admin pass=test"`. scanrunner preserves the raw token rather than joining and re-splitting it.

## 6.3 Output naming

| Target          | Normal report               | XML report                  |
|-----------------|-----------------------------|-----------------------------|
| 10.10.10.10     | results/10.10.10.10.txt     | results/10.10.10.10.xml     |
| dc01.corp.local | results/dc01.corp.local.txt | results/dc01.corp.local.xml |
| 10.10.10.0/24   | results/10.10.10.0_24.txt   | results/10.10.10.0_24.xml   |

## 6.4 Recommended host timeout

```
scanrunner -f targets.txt -sV --host-timeout 10m
```

### Prevent indefinite waits

Without `--host-timeout`, a slow or filtered host can hold a sequential workflow for a long time. Choose a timeout that matches the assessment depth and network conditions.

## 7. Built-in Templates

Templates are optional shortcuts. Extra Nmap arguments supplied after a template are appended to the template arguments.

| Template | Arguments               | Use case                |
|----------|-------------------------|-------------------------|
| quick    | -sV -T4 --top-ports 100 | Fast service inventory. |

| Template    | Arguments                                                     | Use case                                |
|-------------|---------------------------------------------------------------|-----------------------------------------|
| full        | -sS -sV -O -T4 -p-                                            | Full TCP, service, and OS scan.         |
| full-fast   | -sV -A -Pn --min-rate 200 -p-                                 | Faster full-port scan; validate impact. |
| web         | -sV --script http-title,http-headers                          | Web identification.                     |
| web-enum    | -sV --script http-title,http-headers,http-enum                | Expanded web enumeration.               |
| ssl-ciphers | -sV --script ssl-enum-ciphers                                 | TLS cipher review.                      |
| smb-audit   | -sV --script smb-os-discovery,smb-protocols,smb-security-mode | SMB posture review.                     |
| rdp-audit   | -sV --script rdp-enum-encryption,rdp-ntlm-info                | RDP encryption and NTLM information.    |
| udp         | -sU -sV --top-ports 100                                       | Common UDP services.                    |
| vuln        | -sV --script vuln                                             | Broad Nmap vulnerability scripts.       |

## 7.1 Template aliases

```
scanrunner -f targets.txt --profile quick
scanrunner -f targets.txt --template quick
scanrunner -f targets.txt --preset quick
```

## 7.2 Add extra arguments

```
scanrunner -f targets.txt --template full-fast --host-timeout 20m
scanrunner -f targets.txt --template web -p 80,443,8000,8080,8443
```

## 7.3 List exact definitions

```
scanrunner --list-templates
scanrunner --template -vv
```

### Operational caution

The full-fast template includes -Pn, aggressive detection, full ports, and a minimum packet rate. Obtain approval and adjust the rate for fragile or production systems.

## 8. Ping, Discovery, and -Pn

scanrunner has a wrapper ping check for individual targets, while Nmap also has its own host-discovery phase. These are separate decisions.

### 8.1 Default flow

```
Individual IP/hostname
-> scanrunner ping check
-> alive: run Nmap normally
-> no response: prompt whether to continue with -Pn

CIDR target
-> wrapper ping skipped
-> Nmap performs discovery
```

### 8.2 --skip-ping

```
scanrunner -f targets.txt --skip-ping -sV
```

#### Important distinction

--skip-ping skips only scanrunner's wrapper ping. Nmap still performs host discovery unless -Pn is also supplied.

### 8.3 -Pn

```
scanrunner -i 10.10.10.10 -Pn -sV
```

- -Pn tells Nmap to treat the target as online and proceed to port scanning.
- Use it when ICMP or other discovery probes are blocked but the target is known to exist.
- It can increase scan time because every supplied target is treated as up.

### 8.4 Correct capitalization

| Input | Meaning                             |
|-------|-------------------------------------|
| -Pn   | Correct Nmap discovery-bypass flag. |
| -pn   | Incorrect.                          |
| -pN   | Incorrect.                          |
| -PN   | Incorrect.                          |

### 8.5 Why a scan may complete very quickly

- Nmap considered the host down during discovery.

- A narrow port list or top-port scan found no responding ports.
- A hostname failed to resolve.
- A timeout terminated the host quickly.
- The command displayed help or version information instead of scanning.
- A prior complete report was skipped by resume or non-interactive logic.
- The target is localhost and the selected ports are closed.

## 9. Interactive Scan Controls

Sequential interactive mode provides controls for existing reports and live scans.

### 9.1 Existing-report menu

| Key | Action                                        |
|-----|-----------------------------------------------|
| s   | Skip the target.                              |
| r   | Rescan and log it in rescanned.txt.           |
| m   | Mark the target complete without rescanning.  |
| v   | Open the complete report in less or cat.      |
| q   | Quit while preserving existing progress logs. |

### 9.2 Live scan controls

| Key    | Action                                                             |
|--------|--------------------------------------------------------------------|
| Space  | Display target, elapsed time, PID, and open ports observed so far. |
| Ctrl+X | Terminate and skip the current target.                             |
| Ctrl+C | Abort the workflow and preserve progress.                          |

#### TTY requirement

Interactive keyboard controls require a terminal. They are unavailable in parallel mode and may not function when input/output is redirected.

## 10. Non-interactive and Parallel Scanning

--yes enables unattended decisions. --parallel adds bounded concurrency and requires --yes.

## 10.1 Non-interactive mode

```
scanrunner -f targets.txt --yes -sV --host-timeout 10m
```

- Complete existing reports are skipped.
- Incomplete existing reports are rescanned.
- No interactive existing-report menu is shown.

## 10.2 Parallel mode

```
scanrunner -f targets.txt --yes --parallel 4 -Pn -sV
```

- At most N workers run concurrently.
- Interactive keyboard controls are disabled.
- Worker output should be prefixed with the target for readability.
- Each target retains its own report files.

## 10.3 Choosing a worker count

| Environment                    | Suggested starting point | Reason                                                 |
|--------------------------------|--------------------------|--------------------------------------------------------|
| Small lab                      | 2-4                      | Good visibility with modest load.                      |
| Internal LAN                   | 2-5                      | Balance speed and switch/firewall load.                |
| Remote/VPN                     | 1-3                      | Avoid tunnel saturation and packet loss.               |
| Fragile production             | 1                        | Minimize operational impact.                           |
| High-capacity authorized range | 4-10 after testing       | Increase gradually while observing errors and latency. |

## 10.4 Parallel mode with retries

```
scanrunner -f targets.txt --yes --parallel 4 --retries 1 -Pn -sV
```

### Do not confuse speed with completeness

Parallel execution changes elapsed wall-clock time, not the depth of each Nmap command. Confirm the actual Nmap arguments, report completion marker, and inventory.

## 11. Resume, Retry, and Existing Reports

scanrunner uses report contents and audit logs to avoid repeating completed work and to recover from interruptions.

### 11.1 Complete vs incomplete

- A normal report containing the Nmap completion marker is treated as complete.
- A report that exists but lacks the completion marker is incomplete.
- A missing report has not been scanned in the selected output directory.

### 11.2 Resume

```
scanrunner -f targets.txt --resume -sV
```

### 11.3 Retry

```
scanrunner -f targets.txt --retries 2 -sV
```

--retries N means up to N additional attempts after the first failed attempt.

### 11.4 Fresh vs resume prompt

```
[r] Resume (skip completed) [f] Fresh (review all)
```

### 11.5 Output directory isolation

#### Use a unique directory per engagement or scan profile

Resume state is tied to report files in the selected output directory. Mixing unrelated targets or scan profiles in one directory can create confusing skip decisions.

## 12. Scope Enforcement and Exclusions

Scope controls provide a safety boundary before scanner execution. Exclusions modify Nmap targeting but are not a replacement for an engagement allowlist.

### 12.1 Scope file

```
# scope.txt
10.10.0.0/16
192.168.50.10
scanner.example.internal
```

```
scanrunner -f targets.txt --scope-file scope.txt -sV
```

## 12.2 Scope rules

- Exact hostname strings can be allowlisted.
- An individual IP is accepted when contained in an allowlisted CIDR.
- A CIDR target must fit inside an allowlisted network.
- Out-of-scope targets cause refusal before scanning.

## 12.3 Nmap exclusions

```
scanrunner -f targets.txt --exclude 10.10.10.5 --exclude 10.10.10.6 -sV
scanrunner -f targets.txt --exclude-file excluded.txt -sV
```

## 12.4 Scope vs exclusion

| Control        | Purpose                                            | Security posture                     |
|----------------|----------------------------------------------------|--------------------------------------|
| --scope-file   | Positive allowlist: only approved targets may run. | Preferred engagement boundary.       |
| --exclude      | Remove one or more targets from Nmap.              | Convenience filter.                  |
| --exclude-file | Read Nmap exclusions from a file.                  | Convenience filter for larger lists. |

# 13. Reporting, Inventory, and Metadata

scanrunner keeps raw scanner evidence and generates lightweight inventories for review and reporting.

## 13.1 Output selection

```
scanrunner -f targets.txt -o client-network -sV
```

## 13.2 Standard files

```
results/
├─ <target>.txt
├─ <target>.xml
├─ completed.txt
├─ skipped.txt
├─ rescanned.txt
├─ not-pingip.txt
├─ failed.txt
├─ retried.txt
└─ open-ports-inventory.csv
```

```
└─ open-ports-inventory.json
└─ open-ports-report.html # when --html-report is used
```

### 13.3 Audit logs

| File           | Meaning                                                               |
|----------------|-----------------------------------------------------------------------|
| completed.txt  | Targets successfully recorded as complete.                            |
| skipped.txt    | Targets skipped by user or live control.                              |
| rescanned.txt  | Targets selected for rescan.                                          |
| not-pingip.txt | Targets that did not respond to wrapper ping and were skipped/logged. |
| failed.txt     | Scanner failures.                                                     |
| retried.txt    | Targets that required another attempt.                                |

### 13.4 HTML report

```
scanrunner -f targets.txt --html-report -sV
```

### 13.5 Metadata CSV

```
target,owner,environment
10.10.1.25,Platform Team,production
10.10.2.15,IT Operations,development
```

```
scanrunner -f targets.txt --metadata-csv assets.csv --html-report -sV
```

### 13.6 Inventory limitations

#### Open-port parser

The inventory is derived from Nmap normal-output lines matching open ports. Preserve the XML and normal report for full technical evidence, scripts, banners, and host details.

## 14. NetExec (NXC) Workflow

NXC mode runs an installed NetExec protocol and saves raw output plus parsed table exports.



## 14.1 Basic syntax

```
scanrunner -i 10.10.10.10 -nxc smb
scanrunner -f targets.txt -nxc smb
scanrunner -f targets.txt -nxc ldap
scanrunner -f targets.txt -nxc rdp
```

## 14.2 Native options pass-through

```
scanrunner -f targets.txt -nxc smb --timeout 5
scanrunner -f targets.txt -nxc smb -u auditor -p 'Password'
scanrunner -f targets.txt -nxc smb --shares
```

### Credential handling

Avoid placing secrets directly in shell history. Prefer approved secret-handling methods, temporary environment controls, or NetExec-supported files. Protect raw NXC output.

## 14.3 Supported launcher names

The integration may detect either the `nxc` or `netexec` executable name, depending on the installed NetExec version and packaging.

## 14.4 NXC output files

```
results/nxc-smb-YYYYMMDD-HHMMSS-microseconds.txt
results/nxc-smb-YYYYMMDD-HHMMSS-microseconds.csv
results/nxc-smb-YYYYMMDD-HHMMSS-microseconds.json
```

## 14.5 NXC mode restrictions

| Not supported with NXC mode       | Reason                                       |
|-----------------------------------|----------------------------------------------|
| --profile / --template / --preset | Nmap-only arguments.                         |
| --exclude / --exclude-file        | Implemented for Nmap command construction.   |
| --resume                          | Nmap report lifecycle feature.               |
| --skip-ping                       | Nmap wrapper discovery feature.              |
| --retries                         | Nmap scan retry feature in current workflow. |
| --parallel > 1                    | NXC manages its own concurrency.             |
| --metadata-csv / --html-report    | Nmap inventory/report features.              |

## 14.6 NXC help behavior

```
scanrunner -nxc -h
scanrunner -i 10.10.10.10 -nxc smb -h
```

These commands should display scanrunner-focused NXC help and should not start a scan or create empty result files. For full protocol-specific options, use the native NetExec command directly, such as `nxc smb --help`.

## 15. NXC Query Tables

`--nxc-query` selects focused columns parsed from SMB or RDP output. Multiple values may be comma-separated or repeated.

| Query       | Column                                  | Typical protocol |
|-------------|-----------------------------------------|------------------|
| os          | Operating system text                   | SMB              |
| hostname    | Resolved/reported hostname              | SMB/RDP          |
| smbv1       | SMBv1 True/False/None                   | SMB              |
| smb-signing | SMB signing True/False/None             | SMB              |
| null-auth   | Whether the anonymous attempt succeeded | SMB              |
| rdp-nla     | RDP NLA True/False/None                 | RDP              |
| all         | All focused fields                      | SMB/RDP          |

### 15.1 Examples

```
scanrunner -f targets.txt -nxc smb --nxc-query os,hostname,smbv1,smb-signing
scanrunner -f targets.txt -nxc smb --nxc-query null-auth
scanrunner -f targets.txt -nxc rdp --nxc-query rdp-nla
```

### 15.2 Null-auth behavior

When `null-auth` is requested and no `username/password` options are present, scanrunner may add empty credentials for the anonymous test. Confirm this activity is authorized.

## 15.3 Parsing limitations

- Parsing depends on the human-readable NetExec output format.
- A NetExec version change may alter labels or spacing.
- Blank fields do not automatically mean a vulnerability or absence of a feature.
- Use the preserved raw output when validating a table value.

## 16. Target File Splitting

Split mode creates N balanced target files and exits without scanning.

```
scanrunner -f targets.txt --split 3
```

### 16.1 Rules

- N must be at least 1.
- N cannot exceed the count of unique usable targets.
- Blank lines, comments, and duplicates are removed first.
- The original file is unchanged.
- Part files are created beside the source file.

### 16.2 Example distribution

| Unique targets | Parts | Distribution |
|----------------|-------|--------------|
| 10             | 3     | 4, 3, 3      |
| 12             | 4     | 3, 3, 3, 3   |
| 5              | 2     | 3, 2         |

## 17. Shell Completion

The installer can configure tab completion for supported shells. Completion reduces option-typing errors and helps discover templates, protocols, and paths.

### 17.1 Zsh

```
sudo python3 install.py --shell zsh
exec zsh
```

Typical managed configuration:

```
fpath=(~/zsh/completions $fpath)
```

```
autoload -Uz compinit && compinit
```

## 17.2 Bash

```
python3 install.py --shell bash
exec bash
```

## 17.3 Fish and PowerShell

```
python3 install.py --shell fish
python3 install.py --shell powershell
```

## 17.4 Troubleshoot completion

- Confirm the completion file exists under the expected user home, not /root.
- If install.py was run with sudo, verify that SUDO\_USER was respected.
- Reload the shell or start a new session.
- For Zsh, run compinit again after changing fpath.
- Run which scanrunner and confirm the launcher is in PATH.

# 18. Command Cheat Sheets

This chapter is designed for quick copy-and-adjust use during an authorized assessment.

## 18.1 General

```
scanrunner -h
scanrunner -v
scanrunner --help-all
scanrunner --list-templates
```

## 18.2 Nmap quick commands

```
# Top services
scanrunner -f targets.txt --template quick

# Standard service and default-script enumeration
scanrunner -f targets.txt -sV -sC -T4

# Full TCP ports
scanrunner -f targets.txt -Pn -sS -sV -p- -T4 --host-timeout 20m

# Common UDP
scanrunner -f targets.txt --template udp

# Web identification
```

```
scanrunner -f targets.txt --template web

# TLS ciphers
scanrunner -f tls.txt --template ssl-ciphers

# SMB posture
scanrunner -f windows.txt --template smb-audit

# RDP posture
scanrunner -f rdp.txt --template rdp-audit
```

## 18.3 Workflow automation

```
# Resume
scanrunner -f targets.txt --resume -sV

# Non-interactive
scanrunner -f targets.txt --yes -Pn -sV

# Parallel with one retry
scanrunner -f targets.txt --yes --parallel 4 --retries 1 -Pn -sV

# Scope enforced
scanrunner -f targets.txt --scope-file scope.txt -sV

# Reports and metadata
scanrunner -f targets.txt -o client --metadata-csv assets.csv --html-report -sV
```

## 18.4 NetExec

```
# General SMB inventory
scanrunner -f targets.txt -nxc smb

# Focused SMB facts
scanrunner -f targets.txt -nxc smb --nxc-query os,hostname,smbv1,smb-signing

# Anonymous SMB test
scanrunner -f targets.txt -nxc smb --nxc-query null-auth

# RDP NLA
scanrunner -f targets.txt -nxc rdp --nxc-query rdp-nla
```

# 19. Common Assessment Recipes

## 19.1 Fast internal inventory

```
scanrunner -f alive.txt -o internal-inventory --yes --parallel 3 -sV -T4 --top-ports 1000 --host-timeout 8m
```

## 19.2 Full TCP follow-up

```
scanrunner -f priority-hosts.txt -o full-tcp --yes --parallel 2 -Pn -sS -sV -p- -T4 --host-timeout 30m
```

## 19.3 Web-service discovery

```
scanrunner -f web-targets.txt -o web-discovery -sV -p 80,443,8000,8080,8081,8443,9443 --script http-title,http-headers
```

## 19.4 AD/Windows service inventory

```
scanrunner -f windows.txt -o windows-services -Pn -sV -p 53,88,135,139,389,445,464,593,636,3268,3269,3389,5985,5986
```

## 19.5 SMB hardening table

```
scanrunner -f windows.txt -nxc smb --nxc-query os,hostname,smbv1,smb-signing
```

## 19.6 Controlled client engagement

```
scanrunner -f targets.txt --scope-file approved-scope.txt --metadata-csv asset-owners.csv -o client-2026-07 --html-report --yes --parallel 2 --retries 1 -Pn -sV --host-timeout 15m
```

### Recipe principle

Start with a low-impact inventory, review the results, then deepen scans only for approved targets and services.

## 20. Troubleshooting

| Symptom                       | Likely cause                                                                  | Actions                                                                       |
|-------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| scanrunner: command not found | Launcher missing or PATH not refreshed.                                       | Run install.py; check which scanrunner; run hash -r or restart shell.         |
| Permission denied: scanrunner | Target is a directory, script lacks execute permission, or launcher is wrong. | type -a scanrunner; ls -l; chmod +x scanrunner.py; reinstall.                 |
| Version requires -f or -i     | Old script without early version routing.                                     | Run python3 scanrunner.py -v; inspect __version__; update the file and tests. |
| Parallel mode appears silent  | Older quiet worker redirected output.                                         | Update to live target-prefixed worker output; inspect reports and processes.  |
| Hostname appears stuck        | DNS delay/failure.                                                            | getent hosts; dig; verify /etc/hosts; scan the resolved IP.                   |

| Symptom                               | Likely cause                                                                          | Actions                                                                                         |
|---------------------------------------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Scan completes too fast               | Host down, help invoked, prior report skipped, narrow scan, or immediate error.       | Read terminal output; inspect report tail; verify Nmap done; compare with -Pn.                  |
| NXC -h shows huge native help         | -h was forwarded to NetExec.                                                          | Use/update focused scanrunner help; run nxc <protocol> --help only when native help is desired. |
| Empty NXC table                       | No parseable output, target unreachable, protocol mismatch, or changed output format. | Read raw NXC output and exit status.                                                            |
| License/badge images broken in README | External image service unavailable or URL malformed.                                  | Use local repository assets or stable static badges.                                            |
| Completion installed under /root      | Installer run with sudo but wrong home selected.                                      | Update installer to honor SUDO_USER; reinstall for the invoking user.                           |

## 20.1 Inspect a suspiciously fast Nmap result

```
tail -40 results/<target>.txt
grep -n "Nmap done\|Host seems down\|0 hosts up" results/<target>.txt
nmap -Pn -sV <target>
```

## 20.2 Verify the installed code

```
type -a scanrunner
readlink -f "$(which scanrunner)"
sha256sum scanrunner.py
python3 scanrunner.py -v
scanrunner -v
```

## 20.3 Syntax and tests

```
python3 -m py_compile scanrunner.py install.py
python3 -m unittest discover -s tests -v
```

# 21. Testing and Development

The regression suite should use mocked scanner commands and must not send network traffic.

## 21.1 Run tests

```
python3 -m unittest discover -s tests -v
```

## 21.2 Recommended test coverage

- Standalone -v and --version.
- Top-level and nested help routing.
- NXC help must not start a scan or create output.
- Target normalization, comments, and deduplication.
- Hostname-resolution success/error contract.
- Quoted Nmap argument preservation.
- Parallel requires --yes.
- Template catalog.
- NXC SMB/RDP output parsing.
- Completion generation or installer completion paths.
- Scope enforcement.
- Report completion detection.

## 21.3 Pre-release checklist

1. Compile scanrunner.py and install.py.
2. Run all unit tests.
3. Test scanrunner -v from source and installed launcher.
4. Test nested NXC help.
5. Run a mocked end-to-end Nmap scan.
6. Run a mocked end-to-end NXC scan.
7. Install completion as a normal user and through sudo.
8. Review README commands against the actual release.
9. Create and push an annotated Git tag.
10. Publish release notes and mark the intended release latest.

## 22. Complete Option Reference

The following reference covers scanrunner-owned options. Unknown arguments in Nmap mode are passed through to Nmap.

| Option          | Description                                          | Required       |
|-----------------|------------------------------------------------------|----------------|
| -h, --help      | Show top-level or focused help depending on context. | No             |
| -v, --version   | Show scanrunner version when used standalone.        | No             |
| -f, --file FILE | Read targets from a text file.                       | Yes, unless -i |
| -i, --ip TARGET | Use a single target or CIDR.                         | Yes, unless -f |
| --split N       | Split -file into N balanced files and exit.          | No             |
| --profile NAME  | Apply a named Nmap template.                         | No             |



| Option               | Description                                        | Required |
|----------------------|----------------------------------------------------|----------|
| --template NAME      | Alias of --profile.                                | No       |
| --preset NAME        | Alias of --profile.                                | No       |
| --exclude TARGET     | Exclude a target/CIDR; repeatable.                 | No       |
| --exclude-file FILE  | Use a Nmap exclusion file.                         | No       |
| -nxc, --nxc PROTOCOL | Run NetExec protocol mode.                         | No       |
| --nxc-query QUERY    | Select focused NXC table fields.                   | No       |
| --yes                | Run non-interactively.                             | No       |
| --resume             | Skip completed reports.                            | No       |
| --skip-ping          | Skip only the wrapper ping check.                  | No       |
| --no-color           | Disable ANSI colors.                               | No       |
| --scope-file FILE    | Allowlist authorized targets.                      | No       |
| --retries N          | Retry failed Nmap scans N additional times.        | No       |
| --parallel N         | Run up to N Nmap workers; requires --yes when N>1. | No       |
| -o, --output DIR     | Set output directory; default results.             | No       |
| --metadata-csv FILE  | Add target ownership/environment data.             | No       |
| --html-report        | Generate an HTML open-port report.                 | No       |
| --list-templates     | Print template definitions and exit.               | No       |
| --help-all           | Print all focused and option help.                 | No       |

## 22.1 Validation rules

- Exactly one of --file or --ip is required for a scan/split operation.
- --split requires --file and cannot be combined with --nxc.
- --parallel must be at least 1.
- --parallel greater than 1 requires --yes.
- --retries cannot be negative.
- --nxc-query requires --nxc.
- Unknown NXC query values are rejected.
- Nmap-only workflow options are rejected in NXC mode rather than silently ignored.

## 23. File and Data Formats

### 23.1 Target file

```
# Comments and blank lines are permitted
10.10.10.10
10.10.10.20 # database
host.corp.local
10.10.20.0/24
```

### 23.2 Scope file

```
10.10.0.0/16
192.168.1.25
host.corp.local
```

### 23.3 Metadata CSV

```
target,owner,environment
10.10.10.10,Infrastructure,production
host.corp.local,Identity Team,production
```

### 23.4 Open-port inventory CSV

```
target,owner,environment,port_service
10.10.10.10,Infrastructure,production,22/tcp/ssh
10.10.10.10,Infrastructure,production,443/tcp/https
```

### 23.5 Audit log line

```
2026-07-19 18:30:57 | 10.10.10.10
```

### 23.6 NXC JSON

```
[
  {
    "target": "10.10.10.10",
    "hostname": "DC01",
    "os": "Windows Server 2022",
    "smbv1": "False",
    "smb_signing": "True"
  }
]
```

## 24. Operational Best Practices

### 24.1 Before the scan

11. Confirm written authorization, scope, source IP, time window, and emergency contact.
12. Create a dedicated output directory.
13. Validate DNS and routing.
14. Start with low-impact arguments.
15. Use a scope file and exclusions.
16. Record the exact command and scanrunner version.
17. Verify local disk space and Nmap/NetExec versions.

### 24.2 During the scan

- Watch latency, packet loss, client alerts, and system impact.
- Use Space status in interactive mode.
- Do not assume filtered/no-response means secure.
- Reduce worker count or timing when instability appears.
- Keep secrets out of screenshots and shared logs.
- Pause immediately when the client requests it.

### 24.3 After the scan

- Confirm reports contain completion markers.
- Review failed, retried, skipped, and no-ping logs.
- Validate inventory against raw output/XML.
- Archive commands and tool versions.
- Encrypt or securely transfer evidence.
- Remove temporary credentials and sensitive local files.
- Document exclusions and limitations in the assessment report.

### 24.4 Evidence integrity

#### Preserve raw evidence

Do not rely only on summarized CSV/HTML output. Keep original Nmap text/XML and raw NetExec output so findings can be reproduced and reviewed.

## 25. FAQ and Glossary

### Why does scanrunner need Nmap installed?

scanrunner builds and executes Nmap commands; it is not a scanner engine itself.

### Does --skip-ping equal -Pn?

No. --skip-ping skips scanrunner's wrapper ping. -Pn changes Nmap host discovery.

### Why is --parallel restricted to --yes?

Interactive prompts and live keyboard controls cannot safely coordinate across multiple workers.

**Can I pass any Nmap flag?**

In Nmap mode, unknown arguments are passed through. Validate them with `nmap --help`.

**Can I use NetExec modules?**

Native NetExec options are passed through in NXC mode, subject to authorization and installed NetExec support.

**Why is an NXC table blank?**

The raw output may not include parseable facts, the target may be unreachable, or the NetExec output format may differ.

**What does complete mean?**

The report includes the Nmap completion marker; it does not mean every security issue has been found.

**Should I always use -Pn?**

No. Use it when host discovery is blocked or the target is known to be alive. It can make ranges slower.

**Can I scan URLs?**

URL-style input can be normalized to a hostname, but Nmap scans hosts/services, not web paths.

**How should I update?**

Pull a trusted release, run compile/tests, reinstall the launcher, and verify `scanrunner -v`.

**Glossary**

| Term            | Meaning                                             |
|-----------------|-----------------------------------------------------|
| CIDR            | Network-prefix notation such as 10.10.0.0/16.       |
| NSE             | Nmap Scripting Engine.                              |
| NXC             | NetExec command-line tool/mode.                     |
| NLA             | Network Level Authentication for RDP.               |
| SMB signing     | Integrity protection for SMB messages.              |
| Wrapper ping    | scanrunner's preliminary ping check before Nmap.    |
| Host discovery  | Nmap probes used to decide whether a target is up.  |
| Resume          | Skip targets whose reports are already complete.    |
| Scope allowlist | Positive list of targets that are permitted to run. |
| Inventory       | CSV/JSON summary derived from scanner output.       |

## Appendix A - One-Page Quick Reference

```

INSTALL
git clone https://github.com/Madhav-Sai/scanrunner.git
cd scanrunner && python3 install.py

VERIFY
scanrunner -v
scanrunner -h

TARGETS
scanrunner -i 10.10.10.10 -sV
scanrunner -f targets.txt -sV

FULL TCP
scanrunner -f targets.txt -Pn -sS -sV -p- -T4 --host-timeout 20m

PARALLEL
scanrunner -f targets.txt --yes --parallel 4 -Pn -sV

RESUME
scanrunner -f targets.txt --resume -sV

SCOPE
scanrunner -f targets.txt --scope-file scope.txt -sV

REPORTS
scanrunner -f targets.txt -o client --metadata-csv assets.csv --html-report -sV

NXC SMB
scanrunner -f targets.txt -nxc smb
scanrunner -f targets.txt -nxc smb --nxc-query os,hostname,smbv1,smb-signing

HELP
scanrunner --nmap -h
scanrunner --template -h
scanrunner -nxc -h
scanrunner --reports -h
scanrunner --help-all

```

## Appendix B - Recommended Engagement Folder

```

engagement-name/
├── scope/
│   ├── approved-scope.txt
│   ├── exclusions.txt
│   └── assets.csv
├── commands/
│   └── scan-commands.txt
├── results/
│   ├── discovery/
│   ├── full-tcp/
│   ├── udp/
│   └── nxc/
└── evidence/

```

## Appendix C - Manual Revision Information

| Field                | Value                                    |
|----------------------|------------------------------------------|
| Product              | scanrunner                               |
| Product version      | 1.2.0                                    |
| Manual title         | scanrunner v1.2.0 - Detailed User Manual |
| Manual date          | 19 July 2026                             |
| Primary platforms    | Linux, macOS, Windows                    |
| Primary scanner      | Nmap                                     |
| Optional integration | NetExec                                  |
| Intended use         | Authorized security assessments and labs |

**End of Manual**